LEARNING SEMINAR ON DERIVED ALGEBRAIC GEOMETRY

Abstract. The goal of this seminar is to provide an introduction to the theory of Derived Algebraic Geometry. The hope is to make this accessible to Algebraic Geometers who might not be too familiar with the theory of ∞-categories and so we aim to give an introduction to these ideas at the start. Then we will start with the basic notions of Derived Algebraic Geometry and introduce objects of interest such as $\mathrm{QCoh}(X)$ and the cotangent complex $\mathrm{L}_{A/B}$. After we hope to give applications according to interest.

1. Semester Outline

Infinity Categories

1.1. **Introduction & Preliminaries. Speaker:** Carlos
Content: sSet, Infty Cat, homotopy Categories, model structures, homotopy (co)limits ((nice) spaces are generated by the point under homotopy colimits) projective and injective resolutions interpreted via model structures, small object argument,

1.2. **Big Ideas. Speaker:** Carlos
Content: Straightneing-Unstraightening, Yoneda Lemma, ∞-operads, inert/active morphisms, Segal condition, algebras over operads, monads and Barr-Beck

1.3. **Presentable ∞-categories and stability. Speaker:** Carlos
Content: Presentability, adjoint functor theorem, Lurie tensor product, spaces as the unit, stable infinity categories, spectra, triangulated structure, compact and dualizable objects

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1.4. **Derived Commutative Algebras. Speaker: Peter?**
Content: DG categories, Introduce CDGAs (two ways: CAlg(Vect) (and their modules) and what objects are), simplicial commutative rings, animation of rings, model structure on CDGAs (e.g. self-intersubsections), properties of morphisms of rings (etale, smooth, flat, etc.), cotangent complex for rings.

1.5. **Derived Stacks. Speaker:** Ethan
Content: Derived affine schemes and derived schemes, gluings, truncatedness and coconnectiveness, Artin stacks, fibre products (1-coconnective stacks), derived mapping stacks (1-truncated). Serre Intersubsection Formula.

1.6. **Sheaves on Derived Stacks. Speaker:** Ethan
Content: QCoh, gluing examples ($\mathbb{P}^1$, $\mathbb{P}^2$, BG), pullback, pushforward, compact and dualizable objects in QCoh, Ind-categories, IndCoh (example of IndCoh different , functors between QCoh and IndCoh, correspondences and shriek.

1.7. **Cotangent Complex/Derived Deformation Theory. Speaker:** Peter
Content: Construction of cotangent complex. Examples of cotangent complexes eg. BG, Mapping Stacks. Derived deformation theory and derived moduli problems. Connections to Perfect Obstruction Theories.

1.8. **Formal Geometry. Speaker:** Ethan
Content: Formal moduli problems over and under stacks, formal groupoids, quotients by formal groupoids, infinitesimal neighborhoods, Lie algebras and universal enveloping algebra.

Applications

1.9. **Simpson Types? Speaker:** X^{dR}, X^{Betti}, X^{Dol}
Content:

1.10. **Loop Spaces and Connections. Speaker: Peter**
Content:

1.11. **Derived Class Field Theory. Speaker:** Ethan
Content:

1.12. **Perfect Obstruction Theories. Speaker:**
Content: